

Where do videos go to die?

Attractors of long-horizon video generators

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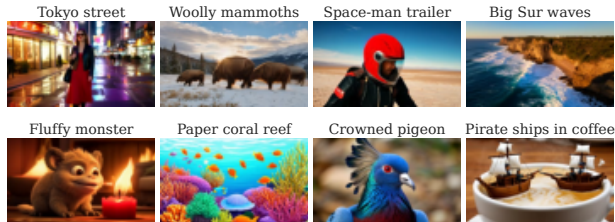
New York University

Research update, October 5, 2026



AI can now generate video that never ends

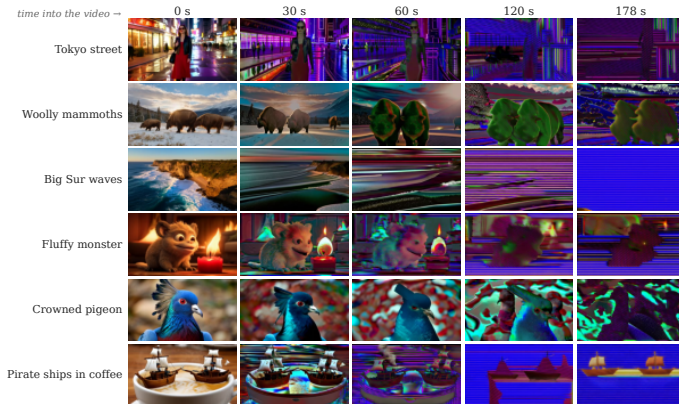
- ▶ New *streaming* video models write a video one short chunk at a time, each chunk looking at the last few.
- ▶ In principle they can run forever: games, live worlds, endless scenes.
- ▶ We ran three of them for **3 minutes** on 8 very different prompts.



The first frame of each of the 8 prompts (Self Forcing).

What happens to a video
if you just keep going?

...but on Self Forcing, they all die the same death

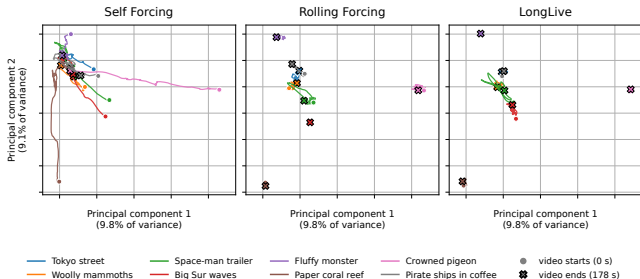


One model (Self Forcing), six prompts, frames from 0 to 178 s. Mammoths, waves, a monster, a pigeon, pirate ships: after three minutes they all become the same purple-blue streaks.

The question, in one sentence

Long videos *decay*. **Where** do they go, **how fast**, and **which design choice** decides it?

Each line is one 3-minute video: its per-second DINOv2 embedding (768-D) projected onto the two directions of largest variation (PCA, fit jointly on all 48 videos; seed 0 shown)



Frame-by-frame trajectories of generated videos, one line per video, colour = prompt. A marble dropped on a landscape rolls into a valley, an *attractor*. Self Forcing's videos all roll into one valley; the other models' videos each stay in their own. The 2-D map keeps 19% of the variation, so slide 13 measures distances in the full embedding.

Why this matters

A common problem, with very different outcomes

Streaming video, game engines and world models generate one step at a time from their own output. Long rollouts often accumulate drift, and recent designs (sinks, long-video tuning) reduce it a lot. In our runs, how a video ends depends strongly on the model and its inference settings.

Measured as symptoms, not as dynamics

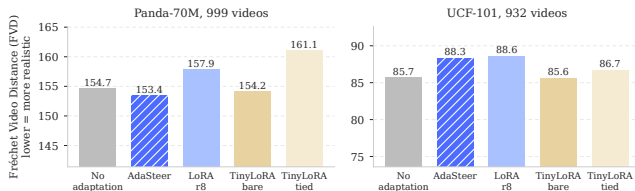
Benchmarks and recent papers already quantify degradation: quality and consistency over long videos (VBench-Long), colour and motion drift, spectral drift, representation collapse, sink collapse. What we have not found is a characterization of the *dynamics*: where trajectories converge, how fast the prompt is lost, how perturbations grow, which design change alters the end state.

What we provide

- ▶ A map of each model's **end states**
- ▶ A **memory half-life** for the prompt
- ▶ **Sensitivity**: does a tiny nudge change the story?
- ▶ **Tipping knobs**: one design change flips the end state
- ▶ Tools anyone can run on a new model

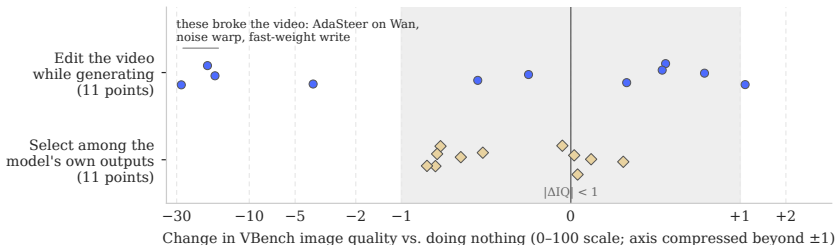
Where we started: fixing videos at test time

- ▶ Original project: adapt the model to each video while generating (*test-time adaptation*, TTA).
- ▶ At 1,000-video scale it does **nothing measurable**: FVD 154.7 → 153.4, within noise.
- ▶ An earlier “+5% FVD” claim did not survive scale. It has been removed from the draft.



LongCat-Video 13.6B, standard-horizon continuation. Lower FVD is better.

33 fixes later: one pattern



Test-time interventions on Wan2.1 Self/Rolling Forcing, each compared with doing nothing on the same clips. Blue: 11 families of methods that *edit* the video as it is made (one point per family, the median of its variants). Tan: 11 results from methods that *pick* among the model's own futures.

Editing the trajectory can wreck a video (down to -28 quality points). Picking among the model's own futures is safe, but only *slows* the decay. So we stopped trying to fix the decay and started measuring it.

Setup

Models (all on Wan2.1, 1.3B)

- ▶ **Self Forcing**: rolling window, no anchor
- ▶ **Rolling Forcing**: keeps the first chunk as an anchor (“sink”)
- ▶ **LongLive**: sink plus long-video tuning

Rollouts

8 MovieGen prompts $\times$ 2 seeds $\times$ 3 models = 48 videos of 3 minutes (2,877 frames each). Plus 128 real 30s clips as a control. About 8 H200-hours so far.

Measurements, every frame

- ▶ **Content**: DINOv2 embedding (what is in the frame)
- ▶ **Look**: brightness, contrast, saturation, colourfulness, sharpness, spectral slope
- ▶ **Motion**: frame difference, optical flow

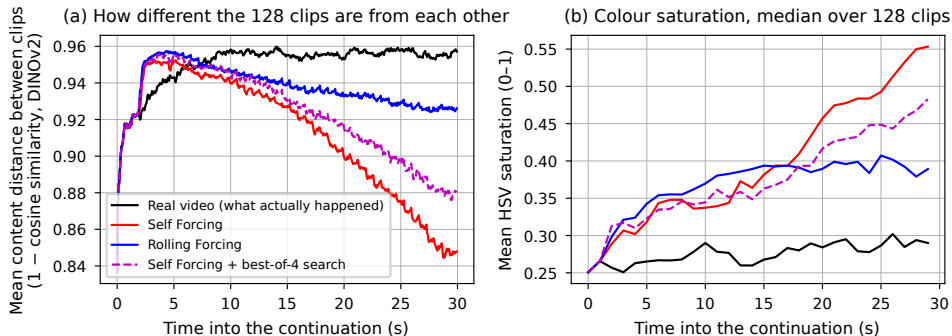
Treat the generator as a dynamical system

The state is the recent frames. The rule is the model. The noise is the random seed.

Toolkit: dynamical-systems questions, video measurements

Concept	Question	Measurement
Attractor	Where do videos end up?	Distance between videos of different prompts; can the prompt still be identified from the frame?
Relaxation	How fast?	Exponential fit to that distance; prompt <i>half-life</i>
Lyapunov (twins)	Does a tiny nudge change the story?	Re-run with one noise block perturbed; track how fast the twins separate
Bifurcation	Which knob flips the end state?	Same weights, change sink size or attention window
Recurrence	Is the motion structured or random?	Recurrence analysis vs. phase-randomized surrogates
Early warning	Can we see collapse coming?	Rising variance and autocorrelation before collapse

Control: real video does not decay



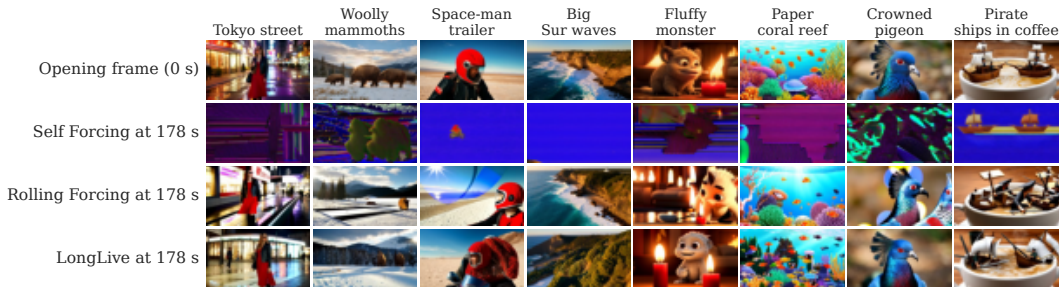
128 Panda-70M clips: each model sees a clip's opening and continues it for 30s; black is what really happened next. (a) Mean content distance between all pairs of the 128 clips at each moment: generated sets become more alike, real video does not. (b) Median colour saturation. Best-of-4 search (purple, dashed) slows the slide but does not stop it.

Three models, three fates



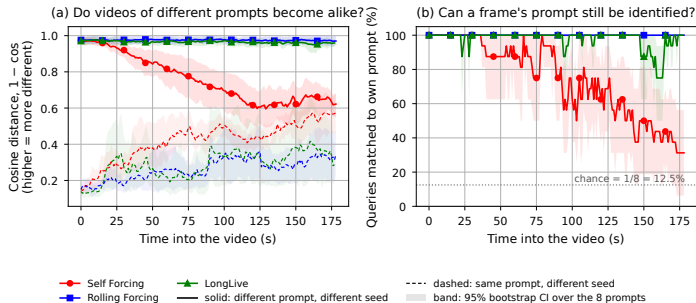
Same prompt, same seed, 0–178 s. Top: “Tokyo street”. Bottom: “Big Sur waves”. Self Forcing collapses, Rolling Forcing holds the scene but whitens, LongLive stays alive.

Where every video ends up



All 8 prompts. Top row: the opening. Below: the frame at 3 minutes for each model.

Self Forcing has one global attractor



(a) Every frame (16/s): resized to 224px short side, centre-cropped 224×224, encoded by DINOv2 ViT-B/14 (dinov2_vitb14); final-layer [CLS] vector, 768-d, averaged per second and normalised. Plotted: mean cosine distance $1 - \cos(z_i, z_j)$ at each second.

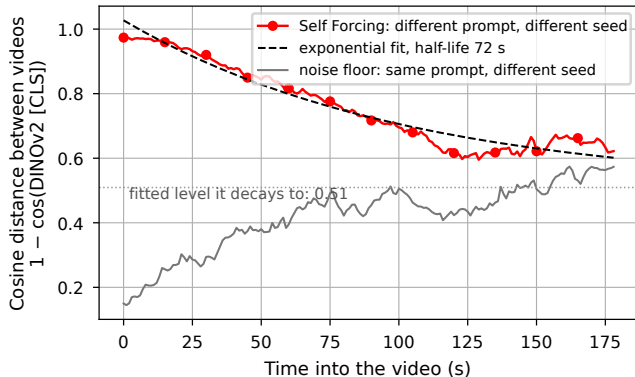
Solid: 56 pairs differing in prompt and seed. Dashed control: 8 pairs with the same prompt, different seed (the spread randomness alone produces).

(b) Nearest-neighbour matching on the same vectors; **no classifier is trained**. Query: one video at second t . Candidates: the 8 videos of the *other* seed at the same second, one per prompt. Correct if the most cosine-similar candidate has the query's prompt. 16 queries per second (8 prompts × 2 seeds). **Chance** = $1/8 = 12.5\%$.

Data, same for all three models: 8 MovieGen prompts × seeds 0 and 1 = 16 videos of 178–179s. Within a seed all prompts get identical noise, so both panels compare only across seeds. Bands: 95% bootstrap over prompts. No smoothing beyond the 1-s average.

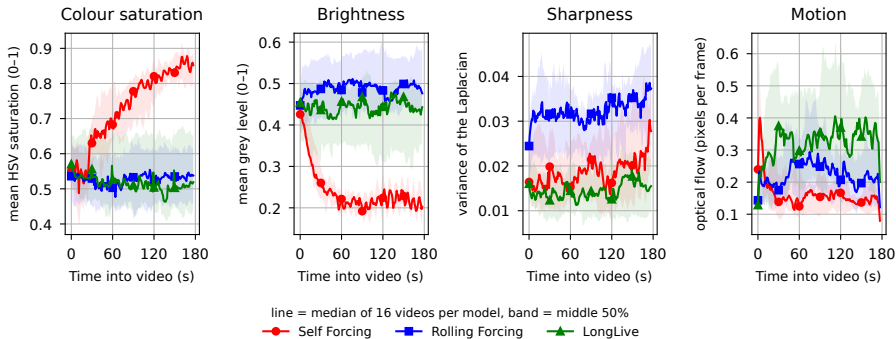
Global attractor: Self Forcing's *different* prompts become nearly as alike as two seeds of the *same* prompt (0.97 → 0.63 vs. 0.57), and a frame stops revealing its prompt (100% → 34%). Rolling Forcing and LongLive keep prompts apart (0.97) and identifiable (100%, 98%).

The prompt has a half-life of about one minute



- Distance between different prompts decays exponentially: half-life **about 1 minute** (58–72 s depending on the fit window and which pairs are averaged).
- It levels off near the **noise floor**: by 3 minutes two different prompts (0.63) are nearly as far apart as two seeds of the *same* prompt (0.57). The prompt is almost fully erased.
- Pull toward the end state is exponential too: $\tau \approx 360$ s, $R^2 = 0.91$.

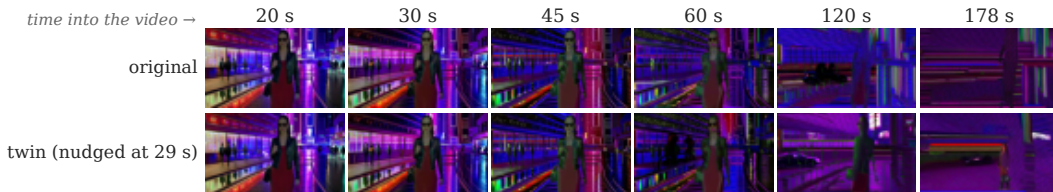
What the decay looks like



Low-level measurements per frame, averaged per second, for the 16 videos of each model (8 prompts $\times$ 2 seeds).

Self Forcing gets darker, more saturated and streaky. Rolling Forcing brightens toward glare. LongLive holds its look. Motion collapses fast (seconds); appearance drifts slowly (minutes).

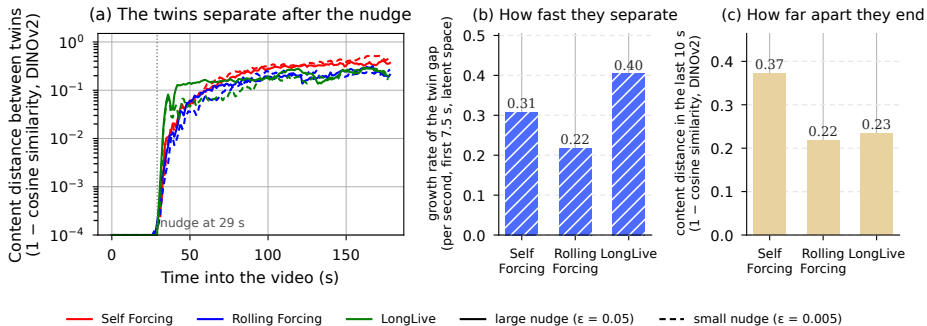
Twins: same video, one tiny nudge



Self Forcing, “Tokyo street”. The runs are bit-for-bit reproducible, so the only difference between the two rows is one block of input noise nudged at 29s. Identical at 20 and 30s, slightly different by 60s, telling different stories by 120s.

Does a tiny nudge change the story? For Self Forcing, yes.

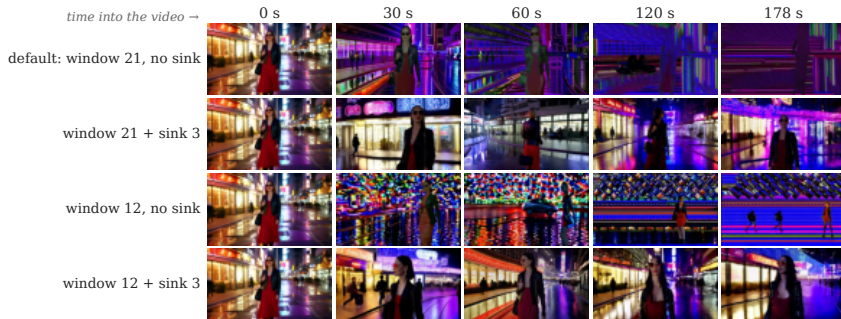
Twins: sensitive is not the same as unstable



All 8 prompts per model, seed 0. ϵ = relative size of the nudge to one noise block. (a) Content distance between twins over time, log scale; the twins are identical before 29s. (b) How fast the gap grows in the model's latent space in the first 7.5s: the same rate for both nudge sizes. (c) Content gap in the last 10s.

LongLive is the *most* sensitive, yet its twins end as close as Rolling Forcing's. Only Self Forcing lets a tiny nudge change the story.

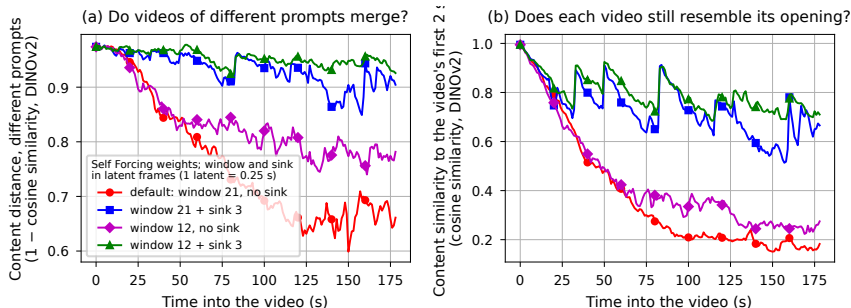
One knob flips the attractor



Same Self Forcing weights, same prompt and seed; only two inference settings change, both counted in latent frames (1 latent frame = 4 video frames = 0.25 s). *Window*: how far back the model attends (21 $\approx$ 5.3 s, 12 = 3 s). *Sink*: the first latent frames kept in memory permanently (3 = the first 0.75 s).

A sink of 3 latent frames keeps the scene alive for 3 minutes. A shorter window alone still collapses, just into a different pattern.

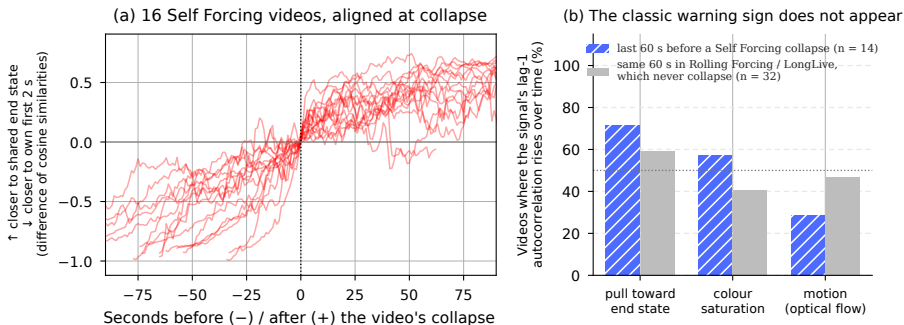
One knob flips the attractor: all 8 prompts



Same four settings, 8 prompts at seed 0 (the 28 prompt pairs, which share the same input noise; on the main runs, same-seed and cross-seed pairs give the same curve, 0.636 vs. 0.628 at 3 min). (a) Mean cosine distance between DINOv2 [CLS] vectors of different prompts. (b) Cosine similarity of each video to its own first 2s.

Without a sink, prompts merge and openings are forgotten (distance $\rightarrow$ 0.67, similarity $\rightarrow$ 0.2).
With a sink, prompts stay apart (>0.9) and openings are kept (≈ 0.7). One knob switches a *global* attractor into *prompt-anchored* ones: a bifurcation.

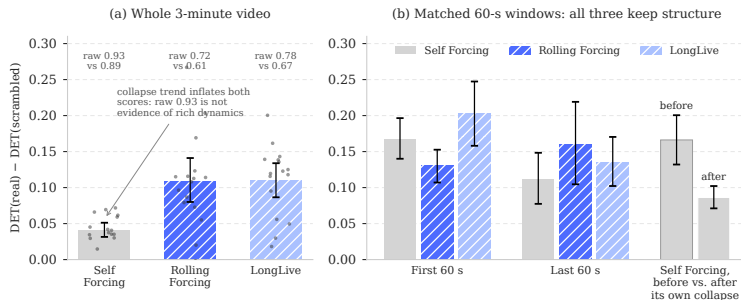
Collapse is a slow ramp, with no early warning



(a) Each line is one Self Forcing video, shifted so its collapse (the first second after which it stays closer to the shared end state than to its own opening for 10s) sits at 0. Going from 10% to 90% of the way takes about 90s. (b) Share of videos whose lag-1 autocorrelation rises during the 60s before collapse (blue) versus the same window in models that never collapse (grey). A real early warning would make blue far higher.

Collapse is a slow ramp, not a tipping point. That is a negative result we will report.

Temporal structure beyond collapse



Per-second DINOv2 [CLS] vectors, 10 principal components per segment; revisit threshold = closest 10% of state pairs; 5 scrambled copies per segment. Collapse time = when a video stays nearer the shared end state than its opening for 10s.

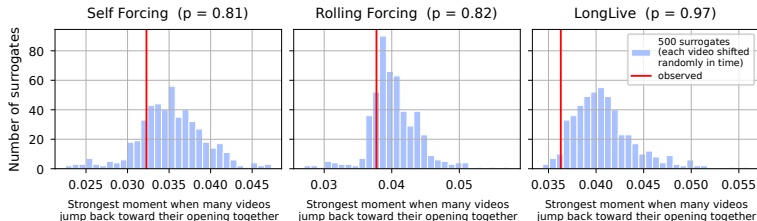
Compare the gap, not the raw score. Over whole videos, Self Forcing's raw 0.93 is inflated by its slow collapse and its gap is only 0.04. In matched 60-s windows all three models keep 0.11–0.20 of structure that scrambling removes, and Self Forcing's gap halves after it collapses (0.17 → 0.09).

Recurrence determinism (DET): share of revisits to an earlier state that continue as a repeated sequence (≥ 3 s), not isolated matches.

Scrambled: phase-randomized copy, same frequency content, timing destroyed.

Plotted: the gap $\text{DET}(\text{real}) - \text{DET}(\text{scrambled})$. Mean of 16 videos, 95% CI; dots = videos.

Checking a rival claim: periodic “snap-back”



LoL (arXiv 2601.16914) reports LongLive snapping back to its first frames at the same moments in every video. For each video we track sudden rises in similarity to its own first second, average them across the 16 videos of a model, and take the strongest moment (red line). Histogram: the same statistic after shifting each video randomly in time, which destroys any shared timing (500 times).

No synchronized snap-back within 3 minutes for any model ($p \geq 0.8$). LoL studies hour-long videos, so a longer LongLive run is still open.

Has anyone done this already?

About 60 targeted searches (2024 to Oct 2026), rechecked Oct 6. Long-horizon degradation is **measured widely**. What we did not find is any work that treats streaming video generators as **dynamical systems**.

Closest work	What it shows	Relation to ours
VBench-Long, Relax Forcing, In-Distribution Forcing (2610.03120)	Quality, consistency, colour and motion drift over minutes	measure symptoms; we model dynamics
LoL (2601.16914)	Sink-collapse loops at hour scale	partial: test longer
Deep / Rolling Sink, FreqForcing (2607.27110)	A sink slows drift; drift as spectral-energy shift	partial: we show a <i>type switch</i>
Effective-rank collapse (2607.27036)	Abrupt representation collapse at drift onset	reconcile with “slow ramp”
Meta-ARVDM (2503.10704)	Theory of history forgetting	frames, not the prompt
Biroli et al., Nat. Commun. 2024	Attractor-like regimes <i>inside</i> the denoising steps	other time axis
Hintze et al., Patterns 2025	Iterated image loops collapse to ~ 12 attractors	concept, not video

Safe claim: existing work measures long-horizon degradation and specific failure modes; we characterize where trajectories converge, how fast prompt information is lost, how perturbations grow, and which inference settings change the attractor. Avoid: “nobody has measured it”, “every model collapses”, “sinks prevent drift” (known).

Next two weeks

Scale and sharpen (about 10 H200-hours)

- ▶ 8 → 32 prompts for every result above
- ▶ Sink sizes 1 and 2: where exactly does the switch happen?
- ▶ Rolling Forcing *without* its sink: does it collapse like Self Forcing?
- ▶ One long LongLive run (beyond 3 min) for the snap-back test

Paper (CVPR 2027, deadline Nov 16)

- ▶ Lead with the attractor map and the one-knob switch
- ▶ Supporting chapters: the TTA null at scale and the 33-intervention atlas
- ▶ Release the toolkit as a drop-in diagnostic

Takeaway. Long videos do not just “get worse”. Each model falls into its own kind of attractor, at a measurable rate, and a single design knob can change which kind.